Circle

Mind map

Geometry facts

- Chord $2r\sin\theta$
- $PA \cdot PB = PT^2$

Equation

- Centre $(-g, -f)$, $r = \sqrt{g^2 + f^2 - c}$
- Image: reflect centre

Diametric form

- $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$
- Smallest circle through A, B

Intercepts

- $2\sqrt{r^2 - p^2}$
- $2\sqrt{g^2 - c}$; touch: $g^2 = c$

Parametric form

- $(h + r\cos\theta, k + r\sin\theta)$
- Max $px + qy$

Point position

- Sign of S_1
- Distances $AC \pm r$

Line and circle

- p vs r
- $c^2 = a^2(1 + m^2)$

Tangent, normal

- $T = 0$; $x\cos\theta + y\sin\theta = a$
- Normal through centre

Tangent length, power

- $L = \sqrt{S_1}$
- Area $PACB = RL$

Director circle

- $x^2 + y^2 = 2a^2$
- Angle 2θ: $CP = r\,\mathrm{cosec}\,\theta$

Mid-point chord

- $T = S_1$
- Locus via CP diameter

Chord of contact, polar

- $T = 0$; $SS_1 = T^2$
- Pole by comparing $T = 0$

Family of circles

- $S_1 + \lambda S_2$, $S + \lambda L$
- $(x - x_1)^2 + (y - y_1)^2 + \lambda L$

Common tangents

- $\sqrt{d^2 - (r_1 \mp r_2)^2}$
- 4, 3, 2, 1, 0 by d

Radical axis

- $S_1 - S_2 = 0$
- Radical centre: equal tangents

Orthogonality

- $2g_1g_2 + 2f_1f_2 = c_1 + c_2$
- $\cos\theta = \frac{r_1^2 + r_2^2 - d^2}{2r_1r_2}$